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Are You Trading Predictably?

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The authors find predictable patterns in stock returns. Stocks whose relative returns are high in a given half hour today exhibit similar outperformance in the same half hour on subsequent days. The effect is stronger at both the beginning and the end of the trading day. These results suggest that strategically shifting the timing of trades can significantly reduce execution costs for institutional traders.

Anecdotal evidence suggests that some institutional traders concentrate trades at particular times of the day. For example, index funds may execute market-on-close orders to minimize tracking error relative to their benchmarks. Active managers may choose to hand trades to the trading desk at particular times of the day. In addition, trading algorithms that imply particular time-of-day trading patterns under certain assumptions have been proposed (e.g., Almgren and Chriss 2000; Grinold and Kahn 2000; Hora 2006). A separate literature suggests that flows of funds to institutional managers are autocorrelated (e.g., Del Guercio and Tkac 2002; Frazzini and Lamont 2008; Cumming and Dai 2009; Lou 2009). Campbell, Ramadorai, and Schwartz (2009) showed that institutional trading is highly persistent—that is, institutional investment managers tend to buy or sell the same stocks on successive days.

These two observations (time-of-day trading patterns and autocorrelated fund flows) suggest the existence of time-of-day patterns in both volume and order imbalances. If these patterns were fully anticipated by traders, however, one would not expect time-of-day patterns in stock prices. The focus of our study was to ascertain whether stock returns exhibit predictable patterns and to measure returns for execution strategies that exploit those patterns.

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Patterns of Resilience in Intraday Stock Returns

Our sample of companies consisted of all NYSE-listed stocks from the post-decimalization period (January 2001–December 2009) that we were able to match with the NYSE Trade and Quote (TAQ) database.[1] The matching yielded data for 4,494 stocks. We used the TAQ database to calculate intraday half-hour stock returns. This approach provided 13 intraday intervals per trading day (from 9:30 a.m. to 4:00 p.m.) and excluded after-hours trading and overnight open/close price movements. In addition to returns, we also measured changes in volume—specifically, the change in the logarithm of the number of shares traded over a half-hour interval. The return and volume data provided measures of price and quantity movements for individual stocks throughout the day.

We analyzed intraday stock returns by using the cross-sectional regression methodology of Jegadeesh (1990). For each lag (k), we ran cross-sectional regressions of half-hour stock returns on returns lagged by k half-hour periods:

$$r_{i,t} = \alpha_{k,t} + \gamma_{k,t} r_{i,t-k} + u_{i,t}, \qquad (1)$$

where $r_{i,t}$ is the return on stock i in the half-hour interval t. The slope coefficients ($\gamma_{k,t}$) represent the response of returns at half-hour interval t to returns over a previous interval lagged by k half-hour periods: the *return responses*. These responses are excess returns on zero net investment portfolios that have returns of 100 percent in a previous half-hour interval (Fama 1976). We calculated the average return response for lag k, $\hat{\gamma}_k$, by averaging the estimated return responses over time $t = 1, \ldots, T$. Note that using cross-sectional regression in this way is different from measuring the autocorrelation of stock returns. In particular, the cross-sectional regression subtracts an overall market effect, which reduces variance and focuses on returns relative to other stocks.

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8. The matrix formulation makes this much clearer

For the five-lag model, define:

$$\mathbf{y} = \begin{bmatrix} r_{1,t} \\ r_{2,t} \\ \vdots \\ r_{N,t} \end{bmatrix}$$

and:

$$X = \begin{bmatrix} 1 & r_{1,t-1} & r_{1,t-2} & r_{1,t-3} & r_{1,t-4} & r_{1,t-5} \\ 1 & r_{2,t-1} & r_{2,t-2} & r_{2,t-3} & r_{2,t-4} & r_{2,t-5} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & r_{N,t-1} & r_{N,t-2} & r_{N,t-3} & r_{N,t-4} & r_{N,t-5} \end{bmatrix}.$$

Then:

$$\mathbf{y} = X\boldsymbol{\beta} + \mathbf{u}$$

where:

$$\boldsymbol{\beta} = \begin{bmatrix} \alpha \\ \gamma_1 \\ \gamma_2 \\ \gamma_3 \\ \gamma_4 \\ \gamma_5 \end{bmatrix}.$$

OLS gives:

$$\boxed{\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'\mathbf{y}}$$

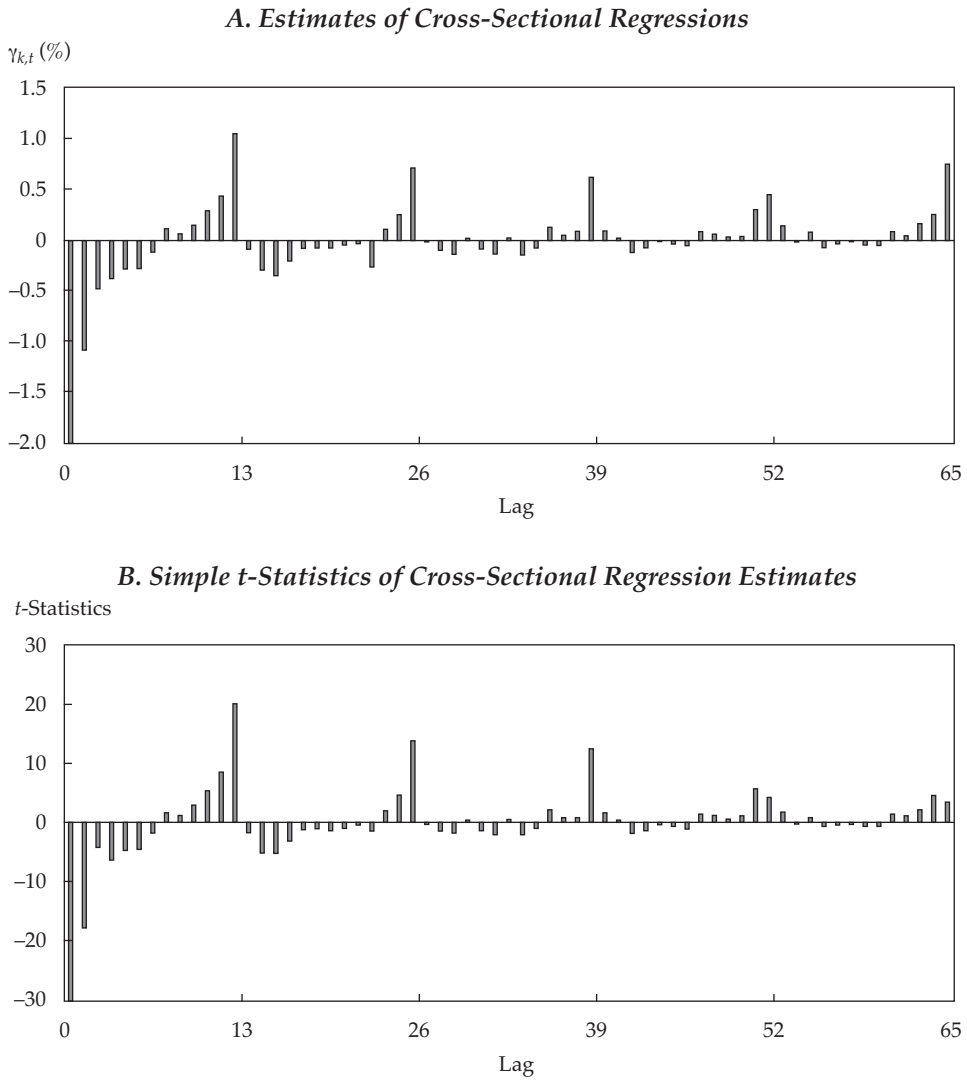
provided $X'X$ is invertible.

This is the mathematical answer to your question.

Figure 1 (Panel A) presents the average return responses for lags up to one week (k = 1, . . ., 65). The data are scaled so that the units for $\hat{\gamma}_k$ are in percentages. The most striking pattern in Figure 1 is that following a period of reversal, the return responses are positive, peaking at horizons that are exact multiples of 13 half hours, or one trading day. Figure 1 clearly shows periodicity at daily intervals through lag 65 (five days). Panel B plots the t-statistics of $\hat{\gamma}_k$, which show the same type of periodicity. Over the first week, the t-statistics at the daily frequency (lags 13, 26, 39, 52, and 65) range from 19.97 at lag 13 to 3.37 at lag 65. **Table 1** reports the numerical results plotted in Figure 1. These results are clearly statistically significant, and their economic significance is an important issue.

The return responses reported in Figure 1 and Table 1, $\hat{\gamma}_k$, are the returns on a zero net investment portfolio whose return k periods ago is 100 percent. Given the implicit leverage in this portfolio, getting a sense of the economic importance of the return periodicity is difficult. For the 2001–05 subperiod, we constructed unlevered zero net investment portfolios by taking long positions in

Figure 1. Cross-Sectional Regression of Returns by Half-Hour Interval, January 2001–December 2009



Notes: Panel A plots the time-series averages of $\gamma_{k,t}$, and Panel B plots the corresponding Fama and MacBeth (1973) t-statistics. We divided the 9:30–16:00 trading day into 13 disjoint half-hour return intervals. For every half-hour interval t and lag k, we ran a simple univariate cross-sectional regression: $r_{i,t} = \alpha_{k,t} + \gamma_{k,t} r_{i,t-k} + u_{i,t}$, where the variable $r_{i,t}$ is the return on stock i in interval t and the variable $r_{i,t-k}$ is the return on stock i in interval $t - k$. We calculated the cross-sectional regression for all combinations of half-hour interval t (29,289 intervals) and lag k (1–65; the past five trading days).

Table 1. Cross-Sectional Regression, January 2001–December 2009

Lag	Estimate	*t*-Statistic	Lag	Estimate	*t*-Statistic
1	–4.74%	–71.52	34	–0.14%	–2.12
2	–1.08	–17.96	35	–0.07	–1.15
3	–0.48	–4.34	36	0.12	2.05
4	–0.38	–6.53	37	0.05	0.66
5	–0.28	–4.85	38	0.08	0.74
6	–0.28	–4.81	39	0.61	12.61
7	–0.12	–1.99	40	0.08	1.61
8	–0.11	1.68	41	0.00	0.04
9	0.06	1.05	42	–0.12	–1.84
10	0.15	2.85	43	–0.08	–1.48
11	0.28	5.23	44	–0.01	–0.16
12	0.43	8.42	45	–0.04	–0.69
13	1.05	19.97	46	–0.06	–1.21
14	–0.09	–1.74	47	0.09	1.54
15	–0.29	–5.33	48	0.05	1.04
16	–0.35	–5.36	49	0.03	0.55
17	–0.20	–3.35	50	0.04	0.87
18	–0.07	–1.30	51	0.29	5.67
19	–0.08	–1.16	52	0.46	4.05
20	–0.08	–1.46	53	0.14	1.81
21	–0.06	–1.17	54	–0.02	–0.24
22	–0.03	–0.60	55	0.08	0.65
23	–0.26	–1.51	56	–0.07	–1.05
24	0.10	2.00	57	–0.04	–0.51
25	0.23	4.69	58	–0.01	–0.18
26	0.70	13.88	59	–0.05	–0.83
27	–0.01	–0.19	60	–0.05	–0.87
28	–0.11	–1.47	61	0.08	1.43
29	–0.14	–1.85	62	0.05	0.89
30	0.01	0.21	63	0.16	2.04
31	–0.09	–1.60	64	0.25	4.43
32	–0.14	–2.25	65	0.74	3.37
33	0.02	0.31			

Notes: This table reports the time-series averages of $\gamma_{k,t}$ and the corresponding Fama and MacBeth (1973) *t*-statistics. We divided the 9:30–16:00 trading day into 13 disjoint half-hour return intervals. For every half-hour interval t and lag k, we ran a simple univariate cross-sectional regression: $r_{i,t} = \alpha_{k,t} + \gamma_{k,t} r_{i,t-k} + u_{i,t}$, where the variable $r_{i,t}$ is the return on stock i in interval t and the variable $r_{i,t-k}$ is the return on stock i in interval $t - k$. We calculated the cross-sectional regression for all combinations of half-hour interval t (29,289 intervals) and lag k (1–65; the past five trading days).

the decile of stocks with the highest returns k periods ago and short positions in the decile of stocks with the lowest returns k periods ago. The average return differential is 3 bps. In the first and last half hours of the trading day, the average return differentials are 11.5 bps and 8.4 bps, respectively. For companies with the smallest (largest) 33 percent of market capitalization, the average return differential is 5.2 bps (1.9 bps), with a *t*-statistic of 16.71 (12.43).

To get a sense of the economic magnitude of this periodicity, let us consider a few comparisons. The ability to capture 3 bps is similar in magnitude to the daily equity risk premium.[2] Indeed, the effect may be larger than the equity risk premium in terms of an incremental Sharpe ratio because one must bear one day of risk to earn the risk premium but need only shift the trade a fraction of a day to take advantage of the periodicity. The ability to capture 3 bps is also similar in magnitude to saving a $0.01 commission on a stock priced at $30 a share (which is in line with institutional commission rates on common stocks). Jeria and Sofianos (2008) evaluated institutional order executions for two alternative algorithms and found a 2.3 bp difference between the implementation

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shortfalls for the average order. They used this result to recommend that traders migrate from one algorithm to the other. Thus, for high-frequency institutional traders, the effect that we found appears to be economically consequential.

The first several return responses are negative, which means that stock returns experience a reversal period lasting several hours. A market is resilient when it replenishes depth rapidly after being depleted by large transactions (see, e.g., Coppejans, Domowitz, and Madhavan 2004; Large 2007). In a resilient market, we expect to see large price impacts owing to large trades, followed by rapid reversals as new limit orders replenish the order book. In less resilient markets, the reversal will take longer, leading to a longer period of negative return responses. The observed negative values of $\hat{\gamma}_k$ at low values of k (see Figure 1 and Table 1) may be due to liquidity shocks when the market is not very resilient. An alternative explanation for the negative values of $\hat{\gamma}_k$ is some combination of bid–ask bounce (e.g., Lehmann 1990; Lo and MacKinlay 1990) and time variation in the frequency of trades at bid versus ask prices (see Keim 1989).

To eliminate the effects of bid–ask bounce, we estimated Equation 1 by using bid-to-bid and ask-to-ask half-hour returns rather than transaction-to-transaction returns. The only significant negative return response in these regressions is for lag $k = 1$, which suggests that temporary liquidity imbalances last less than an hour (two periods). The coefficients for lags 2–12 are positive, most of which are statistically significant. This finding is consistent with intraday momentum, after correcting for bid–ask bounce (Yu, Rentzler, and Wolf 2005). The very significant daily return continuation (lags 13, 26, 39, 52, and 65) persists.

Return Responses by Time of Day

The return patterns in Figure 1 and Table 1 reflect average behavior throughout the day. This effect, however, may be concentrated at certain times of the day. We investigated whether the return responses depend on the time of day. **Table 2** shows the average return responses during half-hour intervals throughout the day. Because the nondaily effect (i.e., at lags other than multiples of 13) is driven mainly by bid–ask bounce, we report the results for only the daily intervals (lags 13, 26, 39, 52, and 65). The return responses are quite pronounced in the first and last half hours of trading. The Day 1 daily response in the opening half hour is 4.19 percent; the response during the last half hour is 2.88 percent. A smaller effect remains during the middle of the day. The Day 1 response at lag 13 is significantly positive in 8 of the 11 midday intervals, averaging 0.60 percent. At longer lags (Days 2–5 in Table 2), the return patterns have a similar shape but are smaller in absolute value. The opening-period return response is significant for Days 2 and 3, whereas the closing-period return response is significant for Days 2–5. When we average intervals 2–12 (i.e., average return responses from 10:00 a.m. to 3:30 p.m.), all the daily return responses are significantly different from zero at conventional levels. This evidence suggests that although the periodicity at daily intervals is stronger at the beginning and the end of the trading day, it exists throughout the day.

The larger effects at the beginning and the end of the day are consistent with the use of certain trading algorithms proposed in the literature (see, e.g., Almgren and Chriss 2000; Grinold and Kahn 2000; Hora 2006). These algorithms are designed to optimize the trade-off between trading costs and the price risk of executing over time. The algorithms in Almgren and Chriss (2000) and Grinold and Kahn (2000) imply that trading rapidly at the beginning of the day and more slowly as time progresses is optimal. The algorithm in Hora (2006) recommends rapid trading at the beginning and the end of the day and slower trading in the middle of the day.

Because of possible incentives to manipulate opening or closing prices, our evidence cannot rule out other explanations for the periodicity (note, however, that our day-of-the-week results, reported later, are inconsistent with closing price manipulation owing to option expiration, as in Ni, Pearson, and Poteshman 2005).

Year-by-Year Analysis

We wished to determine whether our results are driven by particular episodes over the sample period. For example, the general increase in high-frequency trading over the period may alter the pattern over time. **Table 3** reports the average return responses at the daily frequencies, by year. Of the 45 responses reported, 42 are significant, and all the coefficients are significant for Days 1–3. The levels of the return response coefficients are consistent across time, except for a possible reduction in magnitude in 2009.

Patterns in Intraday Volume

Autocorrelated fund flows and a preference for trades at particular times of the day should lead to periodicity in quantity variables (e.g., volume). We examined whether volume has periodicity similar to that of returns and, if so, whether patterns in volume

Table 2. Performance over the Trading Day, January 2001–December 2009

	1 (First)	2	3	4	5	6	7	8	9	10	11	12	13 (Last)	2–12
	9:30–10:00	10:00–10:30	10:30–11:00	11:00–11:30	11:30–12:00	12:00–12:30	12:30–13:00	13:00–13:30	13:30–14:00	14:00–14:30	14:30–15:00	15:00–15:30	15:30–16:00	10:00–15:30
Day 1	4.19%	1.16%	0.94%	0.23%	0.56%	0.66%	0.58%	0.17%	0.42%	0.28%	0.67%	0.95%	2.88%	0.60%
[lag 13]	[15.61]	[6.51]	[4.83]	[1.29]	[3.29]	[4.09]	[3.50]	[0.93]	[2.49]	[1.56]	[3.71]	[5.49]	[12.78]	[11.36]
Day 2	3.05%	0.95%	0.84%	0.28%	0.18%	0.09%	0.34%	0.10%	–0.29%	0.03%	0.25%	0.53%	2.75%	0.30%
[lag 26]	[17.56]	[5.54]	[4.31]	[1.70]	[0.97]	[0.55]	[2.19]	[0.49]	[–1.43]	[0.19]	[1.44]	[3.10]	[13.23]	[5.59]
Day 3	2.43%	0.89%	0.18%	0.41%	0.52%	0.05%	0.37%	–0.18%	0.23%	0.20%	0.26%	0.07%	2.50%	0.27%
[lag 39]	[13.86]	[5.38]	[0.97]	[2.49]	[3.21]	[0.29]	[2.32]	[–0.94]	[1.43]	[1.09]	[1.61]	[0.45]	[11.88]	[5.34]
Day 4	0.53%	0.58%	0.30%	0.48%	0.29%	0.15%	0.08%	0.08%	0.13%	0.51%	0.38%	0.47%	1.96%	0.31%
[lag 52]	[0.39]	[3.53]	[1.64]	[2.86]	[1.78]	[1.05]	[0.58]	[0.50]	[0.78]	[2.94]	[2.25]	[2.85]	[9.40]	[6.31]
Day 5	4.09%	0.77%	0.45%	0.18%	0.50%	0.12%	0.31%	0.57%	0.21%	0.26%	0.27%	0.25%	1.63%	0.35%
[lag 65]	[1.47]	[4.64]	[2.36]	[1.10]	[3.23]	[0.75]	[2.06]	[2.20]	[1.36]	[1.49]	[1.58]	[1.43]	[7.33]	[6.65]

Notes: See notes to Table 1. This table reports the time-series averages of $\gamma_{k,t}$ for each half-hour interval of a trading day and the respective Fama and MacBeth (1973) *t*-statistics (in brackets). Lag k = 13, 26, 39, 52, and 65. There are 2,253 observations for each half-hour interval of a trading day.

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Table 3. Performance by Year, January 2001–December 2009

Year	Day 1 [lag 13]	Day 2 [lag 26]	Day 3 [lag 39]	Day 4 [lag 52]	Day 5 [lag 65]
2001	1.18%	0.75%	0.77%	0.65%	0.51%
	[7.15]	[4.56]	[4.95]	[4.27]	[3.42]
2002	1.01%	0.76%	0.50%	0.61%	0.77%
	[6.31]	[5.07]	[3.42]	[4.28]	[5.33]
2003	1.23%	0.73%	0.46%	0.65%	0.64%
	[9.36]	[5.85]	[3.90]	[5.28]	[5.45]
2004	1.23%	0.83%	0.81%	0.44%	0.56%
	[9.53]	[7.20]	[6.74]	[3.80]	[4.88]
2005	1.31%	0.89%	0.96%	0.55%	0.67%
	[9.15]	[5.85]	[7.10]	[4.01]	[4.87]
2006	0.72%	0.71%	0.61%	0.70%	0.56%
	[4.94]	[5.11]	[4.41]	[5.17]	[3.63]
2007	1.22%	0.80%	0.44%	0.39%	0.50%
	[6.95]	[5.43]	[3.22]	[3.08]	[1.85]
2008	1.20%	0.48%	0.40%	0.62%	1.94%
	[5.98]	[2.58]	[2.23]	[2.69]	[1.02]
2009	0.36%	0.34%	0.53%	–0.48%	0.49%
	[2.27]	[2.03]	[3.26]	[–0.53]	[2.34]

Notes: See notes to Table 1. This table reports the time-series averages of $\gamma_{k,t}$ by year and the corresponding Fama and MacBeth (1973) *t*-statistics (in brackets). Lag k = 13, 26, 39, 52, and 65.

explain the return periodicity. Owing to the persistent nature of volume, we used percentage changes in volume instead of volume itself. Although we found that the percentage change in volume exhibits a pattern similar to the periodic patterns of returns, we concluded that the periodicity in volume does not explain the periodicity in returns.

Using volume data, we repeated the cross-sectional regression:

$$v_{i,t} = a_{k,t} + g_{k,t} v_{i,t-k} + u_{i,t},$$

where $v_{i,t}$ is the percentage change in the volume of stock i traded over the half-hour interval t—that is, $v_{i,t} = \ln(V_{i,t}/V_{i,t-1})$, where $V_{i,t}$ is the volume of stock i in interval t. This transformation arises from the high persistence in the level of volume. We dropped from the sample those periods whose volume was zero. The means and the *t*-statistics of the time series of volume responses are plotted in **Figure 2** (Panels A and B, respectively). The pattern for $\hat{g}_k$ strongly resembles the return response pattern, $\hat{\gamma}_k$, in Figure 1. In particular, the cross-sectional volume response effects are negative except at lags that are multiples of 13.

To study whether the patterns in volume can explain return periodicity, we ran cross-sectional regressions of returns on lagged returns and lagged volume:

$$r_{i,t} = \alpha_{k,t} + \gamma_{k,t} r_{i,t-k} + g_{k,t} v_{i,t-k} + u_{i,t}.$$

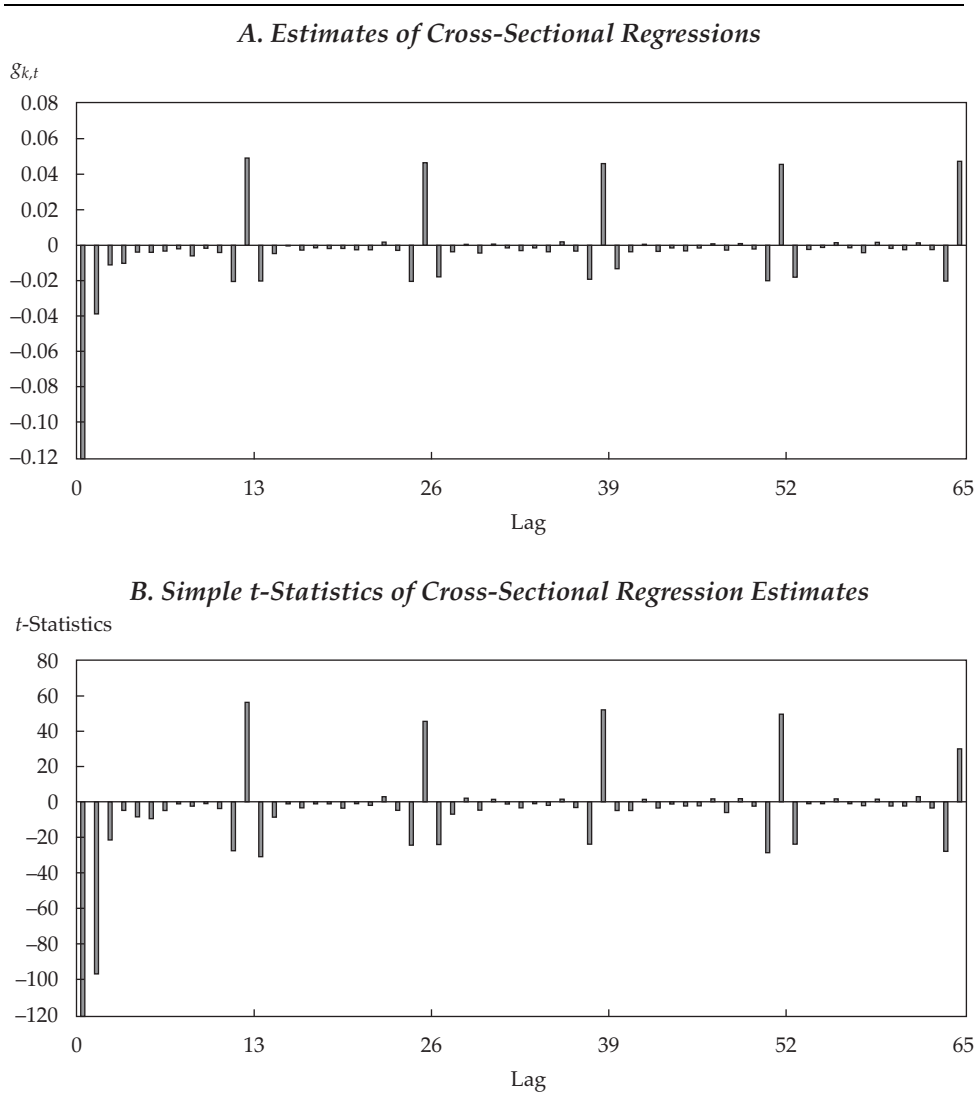
If return predictability is caused primarily by predictability in volume, then including it should diminish or even subsume return predictability based on past returns. Unreported results show that the values of $\hat{\gamma}_k$ remain significant, and the pattern of return responses displayed in Figure 1 remains unchanged. In other words, although volume has the same pattern of predictability as returns, it does not explain predictability in returns.

Robustness

Table 4 reports the daily periodicity for each weekday through lag 65 (five days). The periodicity is similar for all weekdays and is significant for all weekdays through lag 65 except lag 52 for Friday and lag 65 for Monday. **Table 5** shows the daily periodicity by month. The patterns of return responses are similar across all months, with the possible exception of July, which has significant coefficients for lags 13 and 26 but not for lags 39, 52, and 65. The results in Tables 4 and 5 suggest that the periodicity is not driven by a particular day of the week or month of the year.

Heston, Korajczyk, and Sadka (2010) performed a battery of robustness checks for a shorter sample period, 2001–2005. First, although Figure 1 and Table 1 report the results for only five days of lags (k = 13, 26, . . ., 65), the pattern lasts at least 40 trading days (k = 13, 26, . . ., 520). Second, the return patterns are not subsumed by other market activity

Figure 2. Cross-Sectional Regression of Changes in Volume by Half-Hour Interval, January 2006–December 2009



Notes: Panel A plots the time-series averages of $g_{k,t}$, and Panel B plots the corresponding Fama and MacBeth (1973) *t*-statistics. We divided the 9:30–16:00 trading day into 13 disjoint half-hour return intervals. For every half-hour interval t and lag k, we ran a simple univariate cross-sectional regression: $v_{i,t} = a_{k,t} + g_{k,t}v_{i,t-k} + u_{i,t}$, where $v_{i,t}$ is the percentage change in volume (total shares traded during the half-hour interval). For our analysis, we measured volume as the logarithm of the ratio of volume to its prior-one-lag value. We calculated the cross-sectional regression for all combinations of half-hour interval t (13,028 intervals) and lag k (1–65; the past five trading days).

variables, such as order imbalance, volatility, and spreads. Third, the return patterns also exist in pre-decimalization periods in which the minimum price increment was either \$0.125 (1993–1997) or \$0.0625 (1997–2000).

Conclusion

We postulated that systematic trading and institutional fund flows lead to predictable patterns not only in trading volume and order imbalances but also in returns of common stocks. Using half-hour observation intervals for January 2001–December 2009, we studied the periodicity of cross-sectional returns. We found significant continuation of returns at intervals that are multiples of a day.

The pattern does not present a profit opportunity absent other motives to trade because strategies that attempt to take advantage of the daily periodicity lose money, after paying the bid–ask spread. The magnitude of the return pattern, however, is sizable relative to several components of transaction costs, including commissions and

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effective spreads. The return patterns that we documented are consistent with investors' predictable trading patterns. At a minimum, randomizing trades, rather than trading predictably, should help investors avoid buying high and selling low. More strategically, shifting trades to certain periods can substantially reduce execution costs for institutional traders.

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This article qualifies for 1 CE credit.

Table 4. Performance by Weekday, January 2001–December 2009

Weekday	Day 1 [lag 13]	Day 2 [lag 26]	Day 3 [lag 39]	Day 4 [lag 52]	Day 5 [lag 65]
Monday	1.19%	0.66%	0.67%	0.64%	1.77%
	[9.19]	[6.26]	[6.17]	[4.90]	[1.57]
Tuesday	0.85%	0.66%	0.57%	0.59%	0.51%
	[7.88]	[5.78]	[5.52]	[5.47]	[4.97]
Wednesday	1.20%	0.72%	0.61%	0.51%	0.53%
	[10.51]	[6.45]	[5.23]	[4.82]	[4.41]
Thursday	0.98%	0.69%	0.57%	0.51%	0.39%
	[7.77]	[5.95]	[5.32]	[4.67]	[2.34]
Friday	1.06%	0.76%	0.62%	0.04%	0.57%
	[9.42]	[6.68]	[6.09]	[0.08]	[4.43]

Notes: See notes to Table 1. This table reports the time-series averages of $\gamma_{k,t}$ by weekday and the corresponding Fama and MacBeth (1973) *t*-statistics (in brackets). Lag k = 13, 26, 39, 52, and 65.

Table 5. Performance by Month, January 2001–December 2009

Month	Day 1 [lag 13]	Day 2 [lag 26]	Day 3 [lag 39]	Day 4 [lag 52]	Day 5 [lag 65]
January	1.32%	0.74%	0.76%	–0.59%	0.38%
	[7.15]	[4.51]	[4.69]	[–0.46]	[2.30]
February	0.92%	0.78%	0.51%	0.56%	0.70%
	[5.69]	[4.95]	[3.30]	[3.46]	[4.47]
March	1.08%	0.76%	0.88%	0.63%	0.85%
	[6.31]	[4.75]	[5.73]	[4.06]	[5.43]
April	0.92%	0.58%	0.55%	0.76%	0.76%
	[5.54]	[3.56]	[3.53]	[4.87]	[4.92]
May	0.80%	0.94%	0.90%	0.47%	0.42%
	[4.75]	[5.43]	[5.69]	[3.16]	[2.73]
June	0.93%	0.41%	0.88%	0.49%	0.55%
	[5.52]	[2.52]	[5.66]	[2.99]	[3.49]
July	1.24%	0.46%	0.27%	0.29%	–0.27%
	[4.93]	[2.10]	[1.49]	[1.14]	[–0.43]
August	0.92%	0.72%	0.34%	0.52%	0.55%
	[5.37]	[4.43]	[2.22]	[3.43]	[3.67]
September	1.05%	0.74%	0.49%	0.69%	0.63%
	[5.61]	[3.98]	[2.63]	[3.75]	[3.37]
October	1.18%	0.73%	0.56%	0.41%	0.42%
	[6.88]	[4.30]	[3.34]	[2.51]	[2.43]
November	1.07%	0.86%	0.84%	0.40%	0.47%
	[5.26]	[4.97]	[4.73]	[2.35]	[2.80]
December	1.20%	0.69%	0.31%	0.82%	3.44%
	[7.19]	[3.53]	[1.62]	[4.81]	[1.37]

Notes: See notes to Table 1. This table reports the time-series averages of $\gamma_{k,t}$ by month and the corresponding Fama and MacBeth (1973) *t*-statistics (in brackets). Lag k = 13, 26, 39, 52, and 65.

Notes

1. For our study, we expanded the sample period of the analysis in Heston, Korajczyk, and Sadka (2010).
2. An 8 percent equity risk premium corresponds to approximately 3.2 bps per trading day (i.e., 800/250).

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